AJITH E

Software Developer (Python Full Stack)
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SUMMARY

Software developer with **3 years of experience** building web applications and AI-powered tools. I have worked on systems that **track vehicles in real time**, an **AI assistant that answers employee questions** at work, and a **security product that helps companies stop data leaks**. I build both the screens people use (front end) and the behind-the-scenes systems that power them (back end). Currently at Switch Mobility EV, building software that monitors vehicles live and adding AI to make it smarter.

SKILLS (IN PLAIN LANGUAGE)

Screens people use (front end)	React.js, Redux, HTML, CSS, Bootstrap, TanStack
Behind-the-scenes systems (back end)	Python, Django, Django REST Framework (DRF), FastAPI
AI & machine learning	AI assistants using large language models (LLMs); machine-learning models that classify and route requests
Databases (where data is stored)	PostgreSQL, MySQL, AWS Redshift, AWS HBase
Speed-ups (quick-access memory, "caching")	Redis, Memcached, Valkey
Cloud (Amazon Web Services)	S3 (file storage), Lambda (tasks that run automatically), SES (sending emails)
Live updates without refreshing the page	WebSockets, Django Channels
Everyday tools	Git/GitHub (tracking code changes), Docker (packaging apps to run anywhere), Postman (testing APIs)

PROFESSIONAL EXPERIENCE

Senior Officer | Switch Mobility EV — **June 2025 – Present**

- Build and maintain a **web application that tracks vehicles in real time**, processing live data from vehicles so teams can monitor and manage them (Python, Django REST Framework, React.js, Redux).
- Choose the most suitable database for each kind of data so information is stored and retrieved efficiently (PostgreSQL, AWS Redshift, AWS HBase).
- Set up cloud services on AWS for **file storage, tasks that run automatically without managing servers, and automatic email notifications** (S3, Lambda, SES).
- Adding **AI and machine learning** to the existing application to give users smart insights and predictions, automate complex tasks, and support better decisions.

IT Analyst | GI Scripts — **Oct 2024 – May 2025**

- Built and maintained the behind-the-scenes systems for several applications (Python, Django).
- Designed the connections (APIs) that let the screen users see talk to the server smoothly.
- Organized and sped up data storage and retrieval in a MySQL database.
- Wrote automated tests to catch bugs before release, keeping the code reliable.
- Connected applications to outside services (third-party APIs) to add new features.

Associate Software Engineer | Trillion Thoughts Technologies — **May 2023 – Sep 2024**

- Built and maintained interactive web pages (React.js).
- Connected pages to live data sources so users always see up-to-date information.
- Improved the speed and responsiveness of on-screen components.
- Worked closely with back-end developers to improve the overall system.

PROJECTS

AI / Machine Learning

Digital Co-Worker | AI Assistant for Employees

A workplace chatbot that reads an employee's question and sends it to the right place to get answered.

- Built an AI assistant that routes each question to the right source: the company employee directory (Microsoft Entra ID), the IT help desk (ManageEngine), or an AI language model with live web search. A machine-learning model makes the routing decision with **99% accuracy** (FastAPI).
- Designed it so a **new system can be plugged in by adding just one piece of code**.

- Built a **self-improving AI model**: when unsure, it asks the user to pick a category instead of guessing, learns from the answer right away, retrains itself in the background, and swaps in the improved version with **no downtime**, only if accuracy does not drop. No redeployment needed.
- Connected to Microsoft's employee directory (Entra ID, Microsoft Graph) to look up colleagues, reporting managers and teams, and to book meeting rooms.
- Built a reminder service that understands requests written in everyday English and makes sure **the same reminder is never sent twice**, even when several servers are running. Reminders arrive by email (PostgreSQL, AWS SES).

Telematics: Connected Vehicles & Fleet Monitoring

iON Telematic Application | Vehicle Tracking & EV Monitoring Platform

Telematics = technology that collects data from vehicles remotely so it can be tracked and analyzed.

- Developed and maintained a real-time application that collects, processes and displays vehicle data.
- Built the behind-the-scenes APIs (Python, Django REST Framework) and the user-facing screens (React.js, Redux).
- Managed **multiple databases in one application**, automatically choosing the right one for each business need.
- Used Redis caching to keep frequently used data on hand, **reducing database load and speeding up response times**.

Uptime Solution Center (USC) | Fleet Monitoring Platform

A live dashboard for keeping an eye on a fleet of connected vehicles.

- Developed a real-time application for monitoring vehicle status and receiving **live alerts and notifications** from connected vehicles.
- Built fast backend APIs (Python, FastAPI) with a modern web front end (React.js, TanStack).
- Set up **sign-in with existing work credentials**, so employees log in with the same username and password they use on their Windows computer (LDAP).
- Enabled real-time tracking of vehicle status, sensor data and operational events.

Cybersecurity & Data Protection

Watermark Guard | Data Leak Prevention Software (Cybersecurity)

Windows software that places a digital watermark across the screen to discourage and trace leaks of company information.

- **Independently started and built** this product from scratch.
- Built an **admin website** to register devices, manage users and devices, monitor activity, and control watermark settings centrally, protected by secure token-based login.
- Added **live monitoring** showing whether each device is online or offline and whether the app is healthy (WebSockets).
- Built the backend services and APIs for device management, login security, watermark settings, activity tracking and audit records.
- Focused on data protection, security, reliability and accountability (data loss prevention, audit trails).

Web Applications

Hyper Vision / Net Visioner | Network Monitoring Web Application

- Built backend APIs for managing computer networks (Django REST Framework) and designed the database for network layout, performance metrics and event logs.
- Added live monitoring (WebSockets, Django Channels) and tuned APIs and database queries so the system scales as data grows.

Biometric | Employee Attendance Web Application

- Built features for leave management, shift setup and access control.
- Connected the application to biometric attendance devices through APIs to show real-time attendance data, and improved on-screen performance.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science Engineering **CGPA: 8.0**

SKP Engineering College, Tiruvannamalai | 2023